H1 and Artin's conjecture for ρ_{1951} under standard conjectures

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Abstract

ρ is the even icosahedral Artin representation of conductor 1951. This note collects what standard conjectures give for it, and it adds, in this second version, the consequences for the hypothesis H1 of the companion note (automorphy of $\rho|_{G_E}$ over a number field E), a version in which Selberg's conjecture is replaced by Langlands functoriality, and what the ℓ-adic families of the companion work can and cannot give.

- Under the Generalized Riemann Hypothesis every pole of $L(\rho, s)$ lies on the critical line, at a zero of an explicit degree-10 Hecke L-function. GRH cannot do more by itself: an explicit configuration, $6 \otimes \chi^3 - \rho$, is consistent with it.
- Selberg's orthonormality conjecture (Conjecture B) implies that $L(\rho, s)$ is entire; the implication is due to M. R. Murty. Orthonormality among the primitive divisors of two explicit Hecke L-functions, of degrees 12 and 10, already suffices.
- With Booker's theorem this gives modularity of ρ, and with cyclic base change H1 for every Galois extension $E/\mathbb{Q}$ with solvable group. For ρ and its $\sqrt{5}$-conjugate together it gives automorphy of every irreducible representation of $\mathrm{Gal}(N/\mathbb{Q})$, by known functorial lifts.
- The same conclusions follow if the two Hecke characters have automorphic inductions to GL_{12} and GL_{10} over $\mathbb{Q}$, two instances of non-normal automorphic induction. The analytic input is then a theorem (Selberg orthogonality for automorphic L-functions).
- Under the Strong Leopoldt conjecture and the conjectures of Calegari and Mazur, the nearly ordinary families through $\rho|_{G_E}$ do not supply H1.

None of these hypotheses is proved, so nothing here establishes H1 or Artin's conjecture for ρ.

1 Setting

$G = \mathrm{Gal}(N/\mathbb{Q}) \cong \mathrm{SL}(2,5) \times C_5$ and $\rho = 2 \boxtimes \psi^3 = \rho_0 \otimes \chi^3$, where χ is a primitive quintic character mod 1951. The irreducibles of $\mathrm{SL}(2,5)$ are written $1, 2, 2', 3, 3', 4, 4', 5, 6$, with $4 = \mathrm{Sym}^3 \rho_0$ and $4' = \rho_0 \otimes \rho_0'$; $\rho'' = \rho_0' \otimes \chi$ is the $\sqrt{5}$-conjugate of ρ, of conductor 1951. We use three unconditional facts from the companion paper. They are recomputed without GAP in Section 10.

(F1) *Brauer formula.* $\rho = \mathrm{Ind}_{\mathrm{Dic}_5 \times C_5}(\tau_1 \boxtimes \psi^3) - \mathrm{Ind}_{\mathrm{Dic}_3 \times C_5}(\mu \boxtimes \psi^3)$, that is, $L(\rho, s) = L_{12}(s)/L_{10}(s)$. Here L_{12} and L_{10} are Hecke L-functions of finite-order characters over number fields of degrees 12 and 10, with Galois sides $(2 + 4 + 6) \otimes \chi^3$ and $(4 + 6) \otimes \chi^3$. Both are entire.

(F2) *Barrier.* Let $\theta^* = 6 \otimes \chi^3 - \rho$. Then $\langle \theta^*, X \rangle \geq 0$ for every character X induced from a subgroup not containing $\mathrm{SL}(2,5) \times 1$, and for every tensor product of irreducibles outside the twist class of ρ. On the other hand $\langle \theta^*, \rho \rangle = -1$.

(F3) *Monomiality.* $6 \otimes \chi^3 = \mathrm{Ind}_{\mathrm{Dic}_5 \times C_5}(\mu_1 \boxtimes \psi^3)$, so $L(6 \otimes \chi^3, s)$ is an entire Hecke L-function.

We also use Booker's theorem [3]: a 2-dimensional irreducible Artin representation of $G_{\mathbb{Q}}$ whose L-function has only finitely many poles is automorphic. In particular its L-function is then entire.

2 What GRH gives

By GRH we mean that every Hecke L-function of a finite-order Hecke character of a number field has all its nontrivial zeros on $\operatorname{Re} s = \frac{1}{2}$.

Theorem 1 ([T|GRH]). *Assume GRH. Then:*

1. *every pole of $L(\rho, s)$ lies on $\operatorname{Re} s = \frac{1}{2}$, at a zero of L_{10}, and its order is at most the order of that zero;*
2. *either Artin's conjecture holds for ρ, or $L(\rho, s)$ has infinitely many poles on $\operatorname{Re} s = \frac{1}{2}$;*
3. *the number of poles with $0 < \operatorname{Im} s \le T$ is at most $N_{L_{10}}(T) = \frac{10T}{2\pi} \log T + O(T)$.*

Proof. 1. By (F1), $L(\rho, s) = L_{12}(s)/L_{10}(s)$ with L_{12} entire, so a pole of order m at s_0 forces L_{10} to vanish to order at least m at s_0.

- For $\operatorname{Re} s > 1$, L_{10} is a convergent Euler product, so it has no zeros there.
- On $\operatorname{Re} s = 1$ it is nonzero (Hecke).
- For $\operatorname{Re} s < 0$, the zeros of the completed function are excluded by the functional equation. The trivial zeros of L_{10} are cancelled by the poles of its Gamma factors, which are the same as those of L_{12} and $L(\rho)$ on the real axis to the left. Precisely: $\Lambda(\rho) = \Lambda_{12}/\Lambda_{10}$, and the completed functions are entire with all zeros in $0 < \operatorname{Re} s < 1$.

So poles of $\Lambda(\rho)$ lie at nontrivial zeros of L_{10}, which lie on the line by GRH. The Gamma factor of ρ has no zeros, so the poles of $L(\rho)$ in $0 < \operatorname{Re} s < 1$ are those of $\Lambda(\rho)$.

2. If the set of poles is finite, Booker's theorem gives automorphy, hence entireness.

3. This follows from item 1 and the Riemann–von Mangoldt formula for L_{10} (degree 10, conductor 1951^8).

□

3 Why GRH cannot do more by itself

We make precise the statement that no argument that uses only orders of zeros and poles, and their locations, can derive Artin's conjecture for ρ from GRH.

Definition 2. An *order assignment at s_0* is an integer vector $n = (n_\pi)$ indexed by the irreducible characters of G, read as $n_\pi = \operatorname{ord}_{s_0} L(\pi, s)$. It is *admissible* if $\langle n, X \rangle \ge 0$ for every X in the class of (F2).

The true orders at any s_0 form an admissible assignment, because every $L(X, s)$ with X in that class is holomorphic (Hecke, Langlands–Tunnell), and the class is closed under the functoriality listed in (F2).

Proposition 3 ([T]). *Let $s_0 = \frac{1}{2} + i\gamma$ be a zero of $L(6 \otimes \chi^3, s)$. The assignment $n = \theta^*$ is admissible. It has $n_\rho = -1$, $n_{6\otimes\chi^3} = 1$ and all other $n_\pi = 0$. It is compatible with all of the following:*

- *Artin's conjecture for every irreducible outside the twist class of ρ;*
- *holomorphy of every $L(X, s)$ for X in the class of (F2);*

- *GRH for every entire L-function involved, and for $L(\rho)$ in the meromorphic sense (zeros and poles on the line).*

The splitting $e_{6\otimes\chi^3} = \theta^* + e_\rho$, with both summands admissible, places a simple pole of $L(\rho)$ at s_0 and a zero of $L(\rho)$ at a point s_1 on the line, with s_1 as close to s_0 as desired. This leaves the zero counts of every entire L-function attached to G unchanged in any disc containing s_0 and s_1.

Proof. Admissibility is (F2). The assignment involves only the points s_0 and s_1, both on the critical line, so it is compatible with GRH. It changes no order of any $L(X, s)$ with X in the class, beyond what θ^* and e_ρ do separately, and both pair non-negatively with every such X. $L(6 \otimes \chi^3)$ has infinitely many zeros on the line, so the configuration can be repeated at infinitely many heights, consistently with Booker's theorem. □

Remark 4. Proposition 3 is a statement about bookkeeping. It does not construct a counterexample function. What it shows is that GRH moves the question onto the critical line and stops there. A proof of Artin's conjecture for ρ from GRH would need further input about *which* points on the line are zeros of which functions. That is equality information, and it is exactly what numerical enclosures cannot certify (companion paper, Remark 13.5).

4 Artin's conjecture for ρ under Selberg's orthonormality conjecture

The Selberg class. $\mathcal{S}$ is the class of Dirichlet series $F(s) = \sum a_F(n)n^{-s}$ with the following properties:

- they converge absolutely for $\operatorname{Re} s > 1$;
- $(s-1)^m F(s)$ extends to an entire function of finite order for some m;
- they satisfy a functional equation of the standard shape and the Ramanujan bound $a_F(n) \ll n^\varepsilon$;
- they have an Euler product $\log F(s) = \sum b_F(n)n^{-s}$ with b_F supported on prime powers and $b_F(n) \ll n^\vartheta$ for some $\vartheta < \frac{1}{2}$.

Note that $b_F(p) = a_F(p)$. A nonconstant $F \in \mathcal{S}$ is *primitive* if $F = GH$ with $G, H \in \mathcal{S}$ forces $G = 1$ or $H = 1$. Every $F \in \mathcal{S}$ is a finite product of primitive elements [5], because the degree is additive and there are no elements of degree strictly between 0 and 1. Hecke L-functions of finite-order Hecke characters lie in $\mathcal{S}$.

Conjecture B (Selberg). For primitive $F, G \in \mathcal{S}$,

$$\sum_{p \le x} \frac{a_F(p)\overline{a_G(p)}}{p} = \delta_{F,G} \log\log x + O(1).$$

Lemma 5 (unique factorisation under B)**.** *Assume Conjecture B. Let $F_1, \dots, F_r$ be distinct primitive elements and $e_j \in \mathbb{Z}$. If $\prod_j F_j^{e_j} = 1$ as meromorphic functions, then every $e_j = 0$.*

Proof. Taking logarithms and the coefficient of p^{-s} gives $\sum_j e_j a_{F_j}(p) = 0$ for every prime p. Multiply by $\overline{a_{F_k}(p)}/p$ and sum over $p \le x$. Conjecture B gives $e_k \log\log x = O(1)$, so $e_k = 0$. □

Lemma 6 (Chebotarev, unconditional)**.** *For a character ξ of G,*

$$\sum_{p\le x,\ p\nmid 1951} \frac{|\xi(\mathrm{Frob}_p)|^2}{p} = \langle \xi,\xi\rangle \log\log x + O(1).$$

Proof. Write $|\xi|^2 = \sum_C |\xi(C)|^2 \mathbf{1}_C$ over conjugacy classes C. The Chebotarev density theorem with its classical error term gives

$$\sum_{p\le x,\ \mathrm{Frob}_p\in C} \frac{1}{p} = \frac{|C|}{|G|}\log\log x + c_C + o(1),$$

by partial summation. Sum over C and use $\sum_C |C||\xi(C)|^2/|G| = \langle\xi,\xi\rangle$. □

Theorem 7 ([T|Conjecture B])**.** *Assume Selberg's Conjecture B. Then $L(\rho,s)$ is entire. More generally, $L(\pi,s)$ is entire for every nontrivial irreducible π of G, and each such $L(\pi,s)$ is a primitive element of $\mathcal{S}$.*

Proof. Step 1: a single primitive factor. Let π be a nontrivial irreducible character of G. By Brauer's induction theorem, $\pi = \sum_i n_i \operatorname{Ind}_{H_i}\psi_i$ with $n_i\in\mathbb{Z}$ and ψ_i linear, so $L(\pi,s) = \prod_i L(K_i,\psi_i,s)^{n_i}$, a quotient of elements of $\mathcal{S}$. Factor every $L(K_i,\psi_i)$ into primitives and collect the result: $L(\pi,s) = \prod_j F_j^{e_j}$ with distinct primitive F_j and $e_j\in\mathbb{Z}$. Comparing coefficients of p^{-s} in the logarithms gives $\pi(\mathrm{Frob}_p) = \sum_j e_j a_{F_j}(p)$ for $p\nmid 1951$. Lemma 6 with $\langle\pi,\pi\rangle = 1$ gives

$$\log\log x + O(1) = \sum_{p\le x}\frac{|\pi(\mathrm{Frob}_p)|^2}{p} = \sum_{j,k} e_j\overline{e_k}\sum_{p\le x}\frac{a_{F_j}(p)\overline{a_{F_k}(p)}}{p} = \Big(\sum_j e_j^2\Big)\log\log x + O(1),$$

where the last step is Conjecture B. Hence $\sum_j e_j^2 = 1$, so $L(\pi,s) = F_\pi^{\varepsilon_\pi}$ for one primitive F_π and some $\varepsilon_\pi\in\{\pm1\}$.

Step 2: different representations give different factors. If $F_\pi = F_{\pi'}$, then $\varepsilon_\pi\pi(\mathrm{Frob}_p) = \varepsilon_{\pi'}\pi'(\mathrm{Frob}_p)$ for all $p\nmid 1951$. By Chebotarev every conjugacy class is a Frobenius class, so $\pi = \pm\pi'$ as class functions. Evaluating at the identity gives the sign $+$, so $\pi = \pi'$.

Step 3: the sign is $+1$. The Dedekind zeta function satisfies $\zeta_N = \prod_\pi L(\pi,s)^{\pi(1)}$, so $\zeta_N = \zeta\cdot\prod_{\pi\neq 1} F_\pi^{\pi(1)\varepsilon_\pi}$. Since $\zeta_N\in\mathcal{S}$, it is also a product of primitives with non-negative exponents. The Riemann zeta function is primitive in $\mathcal{S}$, and by Step 2 the F_π are distinct; $F_\pi\neq\zeta$ because $\pi\neq 1$, by the argument of Step 2. Lemma 5 shows the two factorisations agree, so $\pi(1)\varepsilon_\pi\ge 0$, that is $\varepsilon_\pi = +1$.

Step 4: entire. So $L(\pi,s) = F_\pi\in\mathcal{S}$, and $(s-1)^m L(\pi,s)$ is entire. Classically, by Brauer and the non-vanishing of Hecke L-functions at $s=1$, $L(\pi,s)$ is holomorphic and non-zero at $s=1$ for nontrivial π. Hence $L(\pi,s)$ is entire. Taking $\pi=\rho$ gives the theorem. □

Remark 8 (how this gets past the barrier)**.** Conjecture B is not a statement about orders of zeros, so Proposition 3 does not apply to it. It controls the prime coefficients of the unknown primitive factors. For ρ these coefficients are known exactly (Frobenius traces), and their mean square is 1 (Lemma 6). In the configuration of Proposition 3 the factorisation of $L(\rho)$ would have to contain a primitive factor to a negative power. Step 3 rules that out using the positivity of the exponents in ζ_N. So the barrier is broken by an input of exactly the kind it predicts: information attached to ρ itself, namely the mean square of its Frobenius traces against the primitive factors of the Hecke L-functions.

Remark 9 (how far Conjecture B is known)**.** Selberg orthogonality is a theorem for cuspidal automorphic L-functions of GL_m in many cases [13]. The difficulty here is that the primitive factors of the Hecke L-functions in Step 1 are not known to be automorphic. Their automorphy is equivalent to Artin's conjecture for the corresponding irreducibles. What is actually needed is far weaker than Conjecture B for all of $\mathcal{S}$: see Section 5, where orthonormality among the primitive divisors of L_{12} and L_{10} (total degree 22) is shown to suffice.

5 A sharper conditional proof: two explicit Hecke L-functions suffice

Theorem 7 uses Conjecture B for all of $\mathcal{S}$. For ρ alone, much less is needed. Step 3 can also be replaced by a degree count, so ζ_N is not needed at all.

Hypothesis B(1951)**.** Let $\mathcal{P}$ be the set of primitive $F \in \mathcal{S}$ that divide L_{12} or L_{10} in $\mathcal{S}$, that is, $L_{12}/F \in \mathcal{S}$ or $L_{10}/F \in \mathcal{S}$. For $F, G \in \mathcal{P}$,

$$\sum_{p \le x} \frac{a_F(p)\overline{a_G(p)}}{p} = \delta_{F,G} \log\log x + O(1).$$

This is Conjecture B restricted to the primitive divisors of two explicit Hecke L-functions of total degree 22. It is a hypothesis about whatever those divisors are. It does *not* reduce to Chebotarev. If Artin's conjecture held for ρ and for $\mathrm{Sym}^3 \rho_0 \otimes \chi^3$, and the three L-functions below were primitive, then $\mathcal{P} = \{L(\rho),\, L(\mathrm{Sym}^3 \rho_0 \otimes \chi^3),\, L(6 \otimes \chi^3)\}$, and B(1951) would follow from Lemma 6 and the orthogonality of characters. Unconditionally, however, Chebotarev gives orthogonality of the *Frobenius traces* only, not membership in $\mathcal{S}$:

- $L(6 \otimes \chi^3)$ lies in $\mathcal{S}$, by (F3), but is not known to be primitive;
- $L(\rho) \in \mathcal{S}$ is the conclusion of Theorem 11;
- $L(\mathrm{Sym}^3 \rho_0 \otimes \chi^3)$ is not known to be entire. The representation $4 \otimes \chi^3$ is not monomial: that would need a subgroup $H \le G$ of index 4, so of order 150. Since $150 \nmid 120$, $H \supseteq 1 \times C_5$ and its image in $\mathrm{SL}(2,5)$ has order 30. That image contains the unique involution $-I$, so its image in A_5 has order 15, and A_5 has no subgroup of order 15. Moreover symmetric-cube functoriality applies to automorphic ρ_0, which is not known.

So the expected three are not known to form $\mathcal{P}$.

Lemma 10 ([T])**.** *There is no $F \in \mathcal{S}$ with $F(s)L(\rho,s) = 1$.*

Proof. Write $\Phi(s) = Q^s \prod_j \Gamma(\lambda_j s + \mu_j) F(s)$ for the completed F, with degree $d_F = 2\sum_j \lambda_j \ge 0$, and $\Lambda(\rho,s) = q^{s/2}\Gamma_{\mathbb{R}}(s)^2 L(\rho,s)$; ρ is even, so both archimedean parameters are 0. Both satisfy functional equations, $\Phi(s) = \omega\overline{\Phi(1-\bar{s})}$ and $\Lambda(\rho,s) = W\Lambda(\bar{\rho}, 1-s)$. Also $\bar{F}(s)L(\bar{\rho},s) = 1$, by conjugating coefficients. If $FL(\rho) = 1$, then the product $\gamma(s) := \Phi(s)\Lambda(\rho,s)$ is a product of exponentials and Gamma factors, of total degree $d = d_F + 2 \ge 2$, and it satisfies $\gamma(s) = \omega W\,\overline{\gamma(1-\bar{s})}$. Stirling's formula gives $\log|\gamma(\sigma+it)| - \log|\gamma(1-\sigma+it)| = d(\sigma - \frac{1}{2})\log|t| + O_\sigma(1)$ as $|t| \to \infty$. The left side is constant, so $d = 0$, a contradiction. □

Theorem 11 ([T|B(1951)])**.** *Assume Hypothesis* B(1951)*. Then $L(\rho,s)$ is entire and primitive in $\mathcal{S}$, so Artin's conjecture holds for ρ.*

Proof. $L_{12}, L_{10} \in \mathcal{S}$ are Hecke L-functions. Each is a finite product of primitive elements [5], all of which lie in $\mathcal{P}$. By (F1), $L(\rho) = L_{12}/L_{10}$. Cancelling equal factors gives $L(\rho, s) = \prod_j F_j^{e_j}$ with distinct $F_j \in \mathcal{P}$ and $e_j \in \mathbb{Z}$. For $p \nmid 1951$ the coefficient of p^{-s} in $\log L(\rho, s)$ is $\rho(\mathrm{Frob}_p)$, so $\rho(\mathrm{Frob}_p) = \sum_j e_j a_{F_j}(p)$. Lemma 6 and B(1951) give, exactly as in Step 1 of Theorem 7, $\log\log x + O(1) = (\sum_j e_j^2) \log\log x + O(1)$. Hence $L(\rho) = F^{\pm 1}$ for one $F \in \mathcal{P}$. The exponent -1 is excluded by Lemma 10. So $L(\rho) = F \in \mathcal{S}$ and $(s-1)^m L(\rho, s)$ is entire. Finally, L_{12} and L_{10} are Hecke L-functions of nontrivial characters, so they are holomorphic and non-zero at $s = 1$. Hence $L(\rho)$ is holomorphic at $s = 1$, and so $L(\rho)$ is entire. Since $L(\rho)$ then has no poles, Booker's theorem [3] also gives the corresponding automorphic form (strong Artin). □

Remark 12 (the barrier again)**.** In the configuration of Proposition 3, $L(\rho)$ has a pole at a zero of $L(6 \otimes \chi^3)$, so $L(\rho)$ is not in $\mathcal{S}$. Then its factorisation $\prod_j F_j^{e_j}$ must contain some $e_j < 0$. Either it is a single F^{-1}, which Lemma 10 excludes, or $\sum_j e_j^2 \geq 2$. B(1951) forbids the second case because ρ's Frobenius traces have mean square 1. So the decisive input is coefficient information about the primitive divisors of L_{12} and L_{10}. Locations of zeros do not enter.

6 From Artin's conjecture to modularity and H1

The companion note uses the hypothesis

> H1(E): there is a cuspidal automorphic representation Π of $\mathrm{GL}_2(\mathbb{A}_E)$ whose local components at almost all places v are unramified with Satake parameter the eigenvalues of $\rho(\mathrm{Frob}_v)$.

Its Theorems A′ and A″ show that H1 for infinitely many imaginary quadratic fields, or for one quadratic and one cyclic cubic field, implies that ρ is modular. The converse direction is the following.

Corollary 13 ([T|B(1951)])**.** *Assume Hypothesis* B(1951)*, for instance Conjecture B. Then:*

1. *ρ is modular: there is a cuspidal automorphic representation π of $\mathrm{GL}_2(\mathbb{A}_\mathbb{Q})$, a Maass form of eigenvalue $\frac{1}{4}$, level* 1951 *and nebentypus χ, with $L(s, \pi) = L(\rho, s)$.*

2. *H1(E) holds for every Galois extension $E/\mathbb{Q}$ with solvable Galois group, with $\Pi = \mathrm{BC}_{E/\mathbb{Q}}(\pi)$; in particular for every quadratic field and every cyclic cubic field. So the hypotheses of Theorems A, A′ and A″ of the companion note hold.*

3. *For such E and every finite-order Hecke character ω of E, $L(s, \rho|_{G_E} \otimes \omega)$ is entire.*

Proof. 1. By Theorem 11, $L(\rho, s)$ is entire, and Booker's theorem [3] gives π.

2. Choose a chain $\mathbb{Q} = E_0 \subset E_1 \subset \cdots \subset E_r = E$ with each E_{i+1}/E_i cyclic of prime degree. Langlands' base change [12] along each step takes a cuspidal representation corresponding to $\rho|_{G_{E_i}}$ to one corresponding to $\rho|_{G_{E_{i+1}}}$, because local base change of unramified representations restricts the Satake parameter. The result is cuspidal unless $\rho|_{G_{E_{i+1}}}$ is reducible. It is irreducible: the Galois group $\mathrm{SL}(2, 5)$ of the field of ρ_0 is perfect, so that field meets E only in $\mathbb{Q}$, and $\rho|_{G_E}$ has projective image A_5.

3. $\Pi \otimes \omega$ is cuspidal, so $L(s, \Pi \otimes \omega)$ is entire. By the companion note (Lemma "compatibility at all places"), Π_v corresponds to $\rho|_{W_{E_v}}$ at every place, so $L(s, \rho|_{G_E} \otimes \omega) = L(s, \Pi \otimes \omega)$.

□

Remark 14. Under Conjecture B for all of $\mathcal{S}$, Murty's theorem holds over every base field [14], so $L(s, \rho|_{G_E} \otimes \omega)$ is entire for every number field E over which ρ stays irreducible and every finite-order ω. Turning this into H1(E) for non-Galois E through the converse theorem over E would also need twists by Hecke characters of infinite order; we do not pursue it, since the companion note only uses quadratic and cyclic fields.

7 Functoriality in place of Selberg's conjecture

Let K_{12} and K_{10} be the fixed fields of $C_{10} \times C_5$ and of $\mathrm{Dic}_3 \times C_5$ in N, of degrees 12 and 10 over $\mathbb{Q}$. Both lie in the A_5-field M: K_{12} is fixed by a subgroup of order 5 of A_5 and K_{10} by a subgroup isomorphic to S_3. By (F1) and Artin reciprocity there are finite-order Hecke characters θ of K_{12} and μ of K_{10} with

$$L_{12}(s) = L(s, \theta), \qquad L_{10}(s) = L(s, \mu), \qquad \mathrm{Ind}_{K_{12}}^{\mathbb{Q}} \theta = (2 + 4 + 6) \otimes \chi^3, \quad \mathrm{Ind}_{K_{10}}^{\mathbb{Q}} \mu = (4 + 6) \otimes \chi^3.$$

Hypothesis AI(1951)**.** There are isobaric automorphic representations Π_{12} of $\mathrm{GL}_{12}(\mathbb{A}_{\mathbb{Q}})$ and Π_{10} of $\mathrm{GL}_{10}(\mathbb{A}_{\mathbb{Q}})$ such that, for almost all primes p, the local L-factors of Π_{12} and Π_{10} at p are those of $L(s, \theta)$ and $L(s, \mu)$. That is, θ and μ have automorphic inductions to $\mathbb{Q}$.

This is a case of Langlands functoriality (automorphic induction along non-normal extensions). It is known for non-normal cubic extensions [7], and Kim [9] obtains non-normal quintic automorphic induction for fields with Galois closure A_5 when the icosahedral representation involved is odd, hence modular. For even icosahedral representations no case is known.

Theorem 15 ([T|AI(1951)])**.** *Assume Hypothesis* AI(1951)*. Then ρ is modular, $L(\rho, s)$ is entire, and the conclusions of Corollary 13 hold.*

Proof. 1. Write $\Pi_{12} = \boxplus_i \sigma_i^{\boxplus a_i}$ and $\Pi_{10} = \boxplus_i \sigma_i^{\boxplus b_i}$ with distinct cuspidal σ_i of $\mathrm{GL}_{n_i}(\mathbb{A}_{\mathbb{Q}})$ and $a_i, b_i \geq 0$. At almost all p the Satake parameters of Π_{12} and Π_{10} are the eigenvalues of $\mathrm{Ind}\, \theta(\mathrm{Frob}_p)$ and $\mathrm{Ind}\, \mu(\mathrm{Frob}_p)$, which are roots of unity. Hence every σ_i is unitary, satisfies the Ramanujan bound at almost all p, and satisfies Hypothesis H of Rudnick and Sarnak (the sums $\sum_p (\log p)^2 |a_{\sigma_i}(p^k)|^2 / p^k$, $k \geq 2$, converge because the coefficients are bounded at almost all p).

2. At almost all p, $\rho(\mathrm{Frob}_p)$ has trace $a_{\Pi_{12}}(p) - a_{\Pi_{10}}(p) = \sum_i e_i a_{\sigma_i}(p)$ with $e_i = a_i - b_i$, by (F1).

3. Selberg orthogonality holds for the σ_i: $\sum_{p \leq x} a_{\sigma_i}(p) \overline{a_{\sigma_j}(p)} / p = \delta_{ij} \log\log x + O(1)$. This is a theorem of Liu, Wang and Ye [13] (unconditional for ranks at most 4, and under Hypothesis H in general), with the self-contragredience condition removed by Avdispahić and Smajlović [2].

4. With Lemma 6, $\log\log x + O(1) = \sum_{p \leq x} |\rho(\mathrm{Frob}_p)|^2 / p = (\sum_i e_i^2) \log\log x + O(1)$. So exactly one e_{i_0} is ± 1 and the others vanish.

5. Comparing degrees, $\sum_i e_i n_i = 12 - 10 = 2$, so $e_{i_0} n_{i_0} = 2$, $e_{i_0} = 1$ and $n_{i_0} = 2$.

6. So $\Pi_{12} \cong \sigma_{i_0} \boxplus \Pi_{10}$. At almost all p the Satake parameter of Π_{12} is the union of those of σ_{i_0} and Π_{10}, so the Satake parameter of σ_{i_0} is the pair of eigenvalues of $\rho(\mathrm{Frob}_p)$. Thus $\pi = \sigma_{i_0}$ corresponds to ρ. By the companion note (last step of the proof of its Theorem A), $L(\rho, s)$ is entire, and the rest is Corollary 13.

□

Remark 16. 1. The proof replaces Hypothesis B(1951), a statement about unknown elements of $\mathcal{S}$, by automorphy of two explicit induced representations, for which orthogonality is known.

2. Conversely, if ρ and ρ'' are modular, then AI(1951) holds: $4\otimes\chi^3$ and $6\otimes\chi^3$ are then automorphic by the symmetric cube and the $\mathrm{GL}_2 \times \mathrm{GL}_3$ functoriality of Kim and Shahidi [11] (Section 8). So AI(1951) is not weaker than what it proves; it is the same statement in the language of functoriality.

3. The two inductions are needed together. AI for θ alone shows that Π_{12} is a sum of three distinct cuspidal representations (the mean square of $\mathrm{Ind}\,\theta$ is 3), but it does not say which of them carries the traces of ρ.

8 Every irreducible representation of G

Corollary 17 ([T|B(1951) for ρ and ρ''])**.** *Suppose that ρ and ρ'' are modular, for instance under Conjecture B, or under Hypothesis* B(1951) *for ρ together with its analogue for ρ''. Then every irreducible representation of $G = \mathrm{Gal}(N/\mathbb{Q})$ of dimension $n \geq 2$ corresponds to a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$.*

Proof. The irreducibles are $X \otimes \psi^j$ with $X \in \{1, 2, 2', 3, 3', 4, 4', 5, 6\}$. Let π_0 and π_0'' correspond to $\rho_0 = \rho \otimes \chi^{-3}$ and ρ_0'.

- 2 and 2′: π_0 and π_0''.
- $3 = \mathrm{Ad}\,\rho_0$ and 3′: the Gelbart–Jacquet lifts [6].
- $4 = \mathrm{Sym}^3\,\rho_0$: the symmetric cube of Kim and Shahidi [11].
- $4' = \rho_0 \otimes \rho_0'$: Ramakrishnan's $\mathrm{GL}_2 \times \mathrm{GL}_2$ product [15].
- $5 = \mathrm{Sym}^4\,\rho_0$: Kim's symmetric fourth power [10].
- $6 = \rho_0 \otimes \mathrm{Ad}\,\rho_0'$: the $\mathrm{GL}_2 \times \mathrm{GL}_3$ product of Kim and Shahidi [11].

Each lift Π is isobaric with the Satake parameters of the corresponding X at almost all p, and twisting by ψ^j is twisting by a Dirichlet character. It is cuspidal: if $\Pi = \boxplus\sigma_i^{\boxplus m_i}$, the Rankin–Selberg L-function $L(s, \Pi \times \widetilde{\Pi})$ has a pole of order $\sum m_i^2$ at $s = 1$ [8]. Up to finitely many Euler factors it is the Artin L-function of $X \otimes \overline{X}$, whose pole at $s = 1$ has order $\langle X, X\rangle = 1$. □

Murty's own reciprocity result under Conjecture B covers solvable Galois groups only [14]; for G, whose quotient $\mathrm{SL}(2, 5)$ is not solvable, the step from entireness to automorphy uses Booker's theorem and the lifts above.

9 The families of the companion work under standard conjectures

Let E be imaginary quadratic and ℓ a prime split in E with $\rho(\mathrm{Frob}_\ell)$ not scalar. The companion note (Proposition "nearly ordinary tangent spaces") distinguishes two choices of ordinary lines at the places $v, \bar{v}$ above ℓ: in case (ii), the base-change choice, the nearly ordinary tangent space has dimension 2 and infinitesimal weights $(w, -w)$; in case (iii), the mixed choice, it has dimension 1 and weight (w, w), $w \neq 0$. Case (iii) is unobstructed, so it carries a one-parameter ℓ-adic family (families/README.md, §1) [T].

- *Strong Leopoldt.* Under the Strong Leopoldt conjecture, Calegari and Mazur [4, Cor. 1.4] show for finite non-dihedral image over an imaginary quadratic field: an infinitesimally classical deformation with diagonal weight forces a twist to descend to an odd representation of $G_{\mathbb{Q}}$, and anti-diagonal weight to an even one. Case (ii) has conditions stable under complex conjugation and anti-diagonal weight, in agreement with ρ being even. In case (iii) complex conjugation exchanges the choice of lines with another one (L_β at v, $\rho(\sigma)L_\alpha$ at $\bar{v}$), so it does not act on the deformation problem and the splitting of the weight space into its eigenlines does not apply. We read the diagonal weight of case (iii) as outside the scope of their corollary. We could not read their §8.2, on which the corollary relies, so this reading is not checked [S].

- *Classical points.* Cuspidal cohomology of Bianchi groups occurs in parallel weight only, so in case (ii) there are no classical points near $\rho|_{G_E}$ other than $\rho|_{G_E}$ itself. In case (iii) the weight is parallel. Calegari and Mazur conjecture that Zariski-dense classical points over fields that are not totally real occur only in exceptional cases, one of which is "even" and, conjecturally, needs finite image [4]. So standard conjectures do not exclude classical points in case (iii). At the first test weight we found none in two models: the odd dihedral form of level 23 at level $23 \cdot 4\mathcal{O}_E$ (no eigen-system modulo 2 with the mixed refinement, computed exactly) and an even tetrahedral representation of conductor 163 at level $163 \cdot 4\mathcal{O}_E$ (324 cuspidal base changes and 8 Eisenstein classes; no other cusp form) [T|A].

- *What the route would still need.* Even with Zariski-dense classical points, H1 would need the automorphy of the classical members, which automorphy lifting over CM fields provides in many ordinary cases [1], and classicality in weight one over E. For the last step no theorem or standard conjecture gives a mechanism: forms of Artin type over E contribute to no cohomology. The Fontaine–Mazur–Langlands conjecture over E contains H1 directly, since $\rho|_{G_E}$ is geometric, and so gives no mechanism either.

So, under standard conjectures, the families route does not supply H1; Sections 6–8 do.

10 Computations

- `groups.py` (output `groups.log`, `groups.json`) recomputes (F1)–(F3) without GAP. $\mathrm{SL}(2,5)$ is $\mathrm{SL}_2(\mathbb{F}_5)$, the characters are built from $\operatorname{tr}\rho_0$ on the nine classes, the irreducible characters of subgroups come from Burnside's algorithm, and every inner product is rounded to an integer after checking that it lies within 10^{-8} of one.
  - (F1): $\mathrm{Ind}_{\mathrm{Dic}_5\times C_5}(\tau\boxtimes\psi^3) = (2+4+6)\psi^3$ and $\mathrm{Ind}_{\mathrm{Dic}_3\times C_5}(\mu\boxtimes\psi^3) = (4+6)\psi^3$; (F3) holds.
  - (F2): θ^* restricts to a genuine character of every subgroup whose image in $\mathrm{SL}(2,5)$ is proper (12 classes of subgroups of $\mathrm{SL}(2,5)$, with the subgroups of G given by Goursat's lemma: 402 pairs of a subgroup and an irreducible character); $\langle\theta^*, X\otimes Y\rangle \geq 0$ for all 1600 products with X, Y outside the twist class of ρ; $\langle\theta^*,\rho\rangle = -1$.
  - Conductor exponents at 1951 (tame inertia generated by an element on which ρ has eigenvalues $1,\zeta_5$): 1 for ρ, 3 for $4\psi^3$, 5 for $6\psi^3$, so 1951^9 for L_{12} and 1951^8 for L_{10}.
  - Inductions from the subgroups $\mathrm{SL}(2,3)\times C_5$ of index 5 never separate $2\psi^j$ from $2'\psi^j$, so no expression of $L(\rho)$ through index-5 inductions exists; degree 12 is the least for the numerator.

- `orth.py` (output `orth.log`, `orth.json`) evaluates the prime sums of Lemma 6 with the Frobenius classes at all 23 350 552 primes below $4\cdot 10^8$ [N]. The differences $S_{XY}(x) - \langle X,Y\rangle \log\log x$ are stable to 10^{-3} from $x = 10^5$ on:

x	(ρ,ρ)	$(\rho,4\psi^3)$	$(\rho,6\psi^3)$	(L_{12},L_{12})	(L_{10},L_{10})
10^5	0.0302	0.5619	−0.8587	−4.1397	−3.5765
10^7	0.0295	0.5612	−0.8590	−4.1480	−3.5820
$4\cdot 10^8$	0.0294	0.5611	−0.8592	−4.1507	−3.5840

These are the unconditional half of Theorems 11 and 15. Hypothesis B(1951) concerns the unknown primitive divisors of L_{12} and L_{10}, and sums of this kind cannot test it.

11 Summary

hypothesis	$L(\rho,s)$ entire	ρ modular	H1(E), $E/\mathbb{Q}$ solvable Galois	all irreducibles of G
none	poles only at zeros of L_{10}, with $\delta(t) < \operatorname{Re} s < 1-\delta(t)$, $\delta(t) = (8912 + 1145\log\lvert t\rvert)^{-1}$, $\lvert t\rvert \geq 1$	—	—	—
GRH	poles only on $\operatorname{Re} s = \frac{1}{2}$ at zeros of L_{10}	—	—	—
GRH, Artin for all other irreducibles, known functoriality	not enough (Proposition 3)	—	—	—
B(1951)	yes	yes	yes	with B(1951)″: yes
Conjecture B	yes	yes	yes	yes
AI(1951)	yes	yes	yes	with its analogue for ρ'': yes
H1 for one quadratic and one cyclic cubic field, or for infinitely many imaginary quadratic fields (companion note)	yes	yes	yes	with the same for ρ'': yes
Strong Leopoldt, Calegari–Mazur, Fontaine–Mazur (families)	no conclusion	—	—	—
Langlands reciprocity, or Fontaine–Mazur–Langlands over $\mathbb{Q}$ or E	yes (contains the statement)	yes	yes	yes

The first non-trivial row is Conjecture B, through Murty's argument; the most economical is B(1951), about two explicit L-functions. None of the hypotheses in the table is proved. The "none" row uses the zero-free region of Lee recorded in the first version.

References

[1] P. B. Allen, F. Calegari, A. Caraiani, T. Gee, D. Helm, B. V. Le Hung, J. Newton, P. Scholze, R. Taylor and J. A. Thorne, *Potential automorphy over CM fields*, Ann. of Math. (2) 197 (2023).

[2] M. Avdispahić and L. Smajlović, *On the Selberg orthogonality for automorphic L-functions*, Arch. Math. 94 (2010), 147–154.

[3] A. R. Booker, *Poles of Artin L-functions and the strong Artin conjecture*, Ann. of Math. 158 (2003).

[4] F. Calegari and B. Mazur, *Nearly ordinary Galois deformations over arbitrary number fields*, J. Inst. Math. Jussieu 8 (2009), 99–177.

[5] J. B. Conrey, A. Ghosh, *On the Selberg class of Dirichlet series: small degrees*, Duke Math. J. 72 (1993).

[6] S. Gelbart and H. Jacquet, *A relation between automorphic representations of* GL(2) *and* GL(3), Ann. Sci. École Norm. Sup. (4) 11 (1978).

[7] H. Jacquet, I. I. Piatetski-Shapiro and J. Shalika, *Relèvement cubique non normal*, C. R. Acad. Sci. Paris 292 (1981).

[8] H. Jacquet and J. A. Shalika, *On Euler products and the classification of automorphic representations I, II*, Amer. J. Math. 103 (1981).

[9] H. H. Kim, *An example of non-normal quintic automorphic induction and modularity of symmetric powers of cusp forms of icosahedral type*, Invent. Math. 156 (2004).

[10] H. H. Kim, *Functoriality for the exterior square of* GL_4 *and the symmetric fourth of* GL_2, J. Amer. Math. Soc. 16 (2003).

[11] H. H. Kim and F. Shahidi, *Functorial products for* $\mathrm{GL}_2 \times \mathrm{GL}_3$ *and the symmetric cube for* GL_2, Ann. of Math. (2) 155 (2002).

[12] R. P. Langlands, *Base change for* GL(2), Ann. of Math. Stud. 96, Princeton, 1980.

[13] J. Liu, Y. Wang and Y. Ye, *A proof of Selberg's orthogonality for automorphic L-functions*, Manuscripta Math. 118 (2005), 135–149.

[14] M. R. Murty, *Selberg's conjectures and Artin L-functions*, Bull. Amer. Math. Soc. 31 (1994), 1–14.

[15] D. Ramakrishnan, *Modularity of the Rankin–Selberg L-series, and multiplicity one for* SL(2), Ann. of Math. (2) 152 (2000).